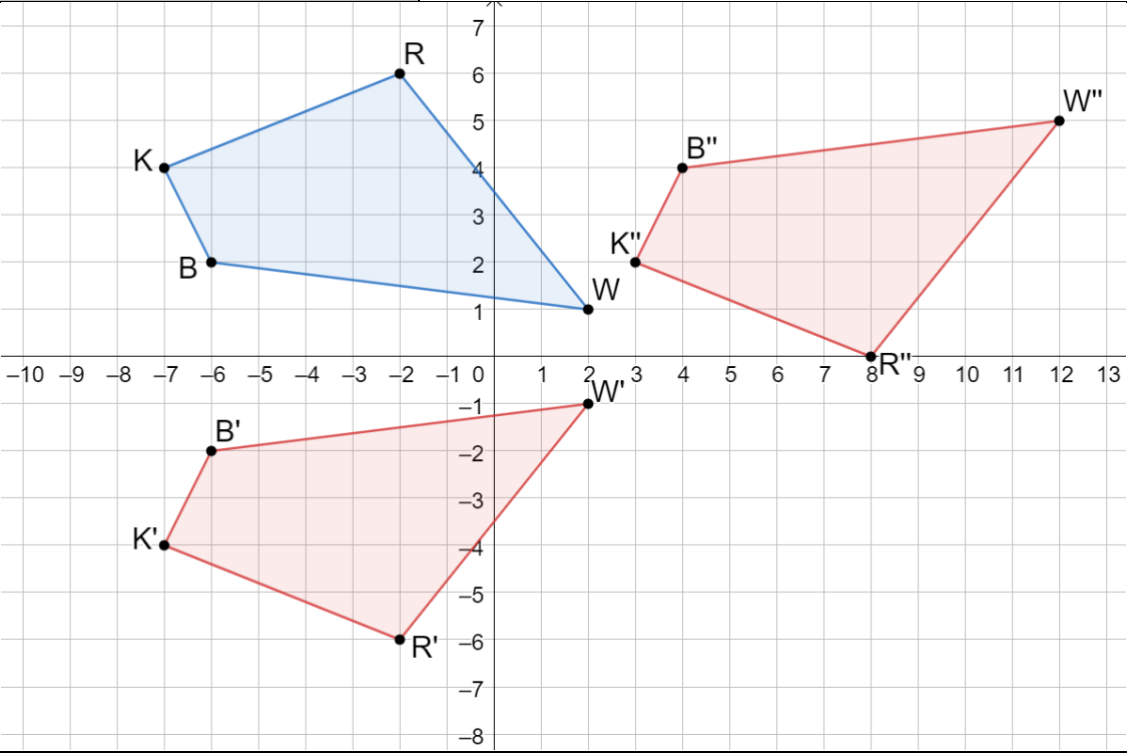
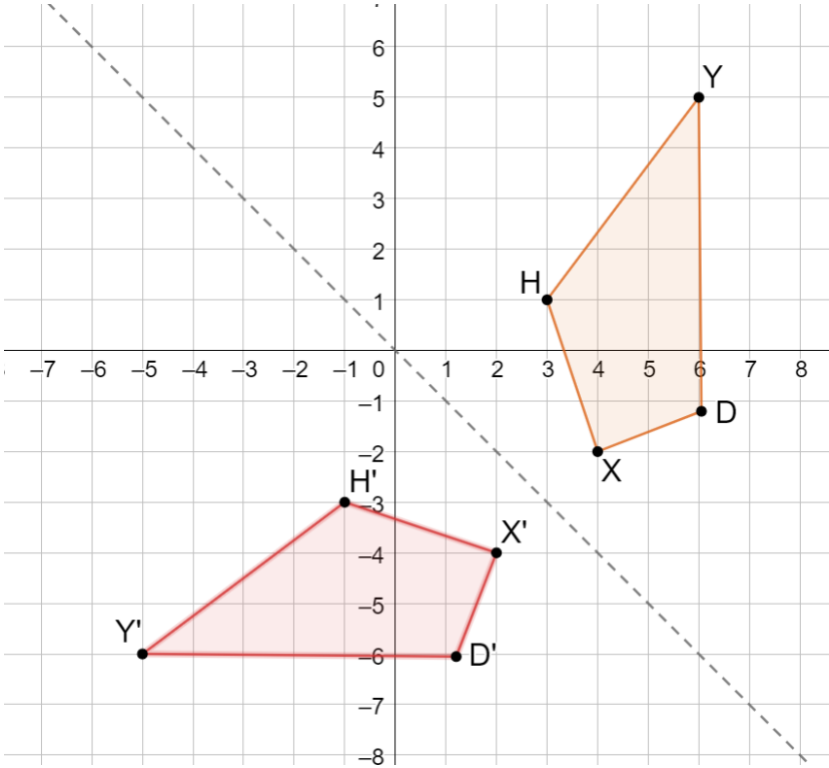


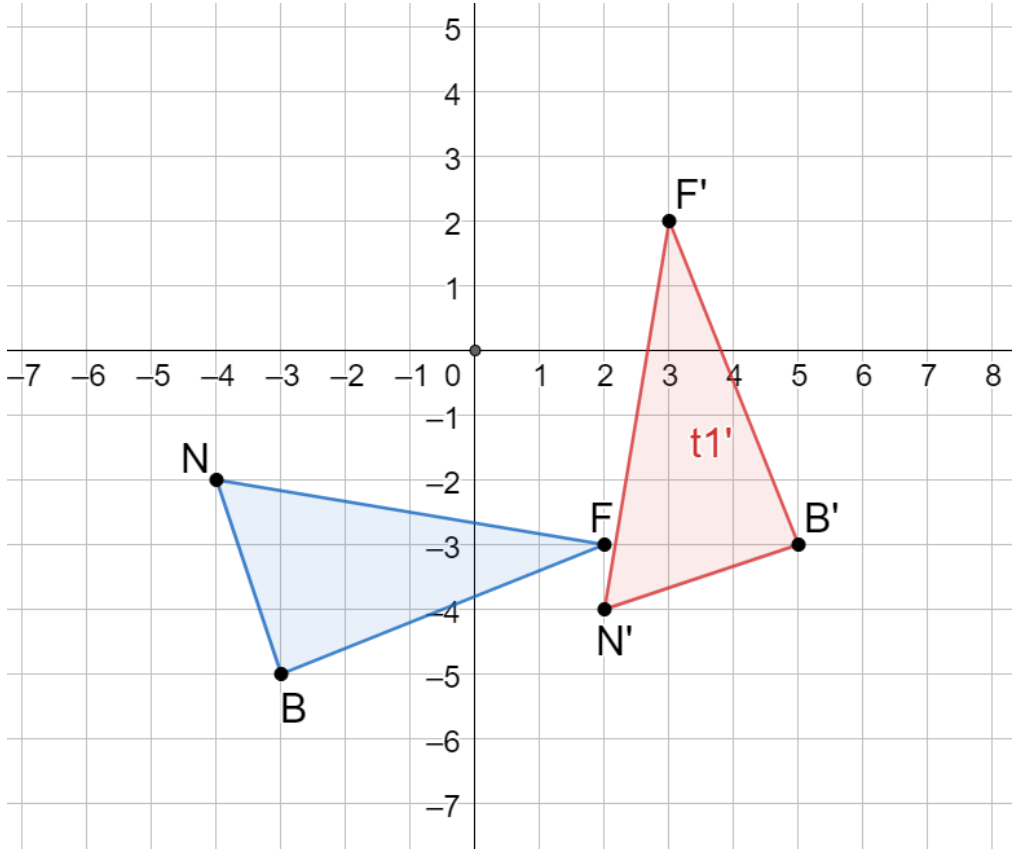
Geometry Unit 4 Assessment Guide

General Information	The benchmarks in this unit have similarities. For some of the questions on the assessment the question is asking the student to show knowledge on content that is shared between more than one benchmark, as noted. The nuances of the benchmarks can be found in the language of each benchmark. For example, MA.912.GR.2.1 states "Given a preimage and image" which means students should not be creating the image. Whereas, MA.912.GR.2.5 states "Given a geometric figure and a sequence of transformations" which means that the preimage and a sequence of transformations should be given for the student to create the image.
Calculator Information	All questions are calculator neutral. Students may be given the use of a calculator and it will not have influence on their ability to show their knowledge of the content.
Total Recommended Points	The total recommended points in this guide is 92 points. Teachers can either determine where best to use the extra 8 points or round the total value up to 100 points.

Question	1a	Benchmark(s)	MA.912.GR.2.5
		Recommended Scoring (8 Total Points)	Suggested points are 3 points for each transformation and 1 point for correctly labeling each image.
		Additional Notes	If a student incorrectly draws the result of the first transformation but correctly applies the second transformation to their incorrect transformation then the student has correctly completed 4 out of the 8 points.

Question	1b	Benchmark(s)	MA.912.GR.2.2, MA.912.GR.2.6
<i>Answers will vary. Possible responses are shown.</i> <i>The first transformation is a reflection and the second transformation is a translation. Both of these transformations are rigid motions which preserve both distance and angle measure.</i> <i>All points on quadrilateral $KRWB$ can be mapped to all points on quadrilateral $K'R'W'B'$ because $(x,y) \rightarrow (x,-y)$ is a rigid motion. All points on quadrilateral $K'R'W'B'$ can be mapped to all points on quadrilateral $K''R''W''B''$ because $(x,y) \rightarrow (x + 10,y + 6)$ is a rigid motion.</i>	Recommended Scoring (4 Total Points)	Student responses will vary.	
	Additional Notes	Consider partial credit for students who correctly note rigid motions, distance is preserved, or definition of congruence in terms of rigid motions but also include incorrect information.	

Question	2	Benchmark(s)	MA.912.GR.2.5
		Recommended Scoring (4 Total Points)	The question does not state that the student should label the result. Use your discretion and class instruction to determine partial credit if a student does not label the result.

Question	3	Benchmark(s)	MA.912.GR.2.5
		Recommended Scoring (4 Total Points)	The question does not state that the student should label the result. Use your discretion and class instruction to determine partial credit if a student does not label the result.

Question		4	Benchmark(s)	MA.912.GR.2.1, MA.912.GR.2.3
	Quadrilateral $RKHD$ Transformed to Quadrilateral $R'K'H'D'$	Quadrilateral $R'K'H'D'$ Transformed to Quadrilateral $R''K''H''D''$	Recommended Scoring (8 Total Points)	Suggested points are 4 points for each description.
Written Description	<i>Quadrilateral $RKHD$ is translated left 4 units and up 8 units to create quadrilateral $R'K'H'D'$.</i>	<i>Quadrilateral $R'K'H'D'$ is rotated 180° clockwise about the origin to create quadrilateral $R''K''H''D''$.</i>	Additional Notes	Students may include more than one transformation for each description. Carefully read each response to consider if the student's response could work.

Question	5a	Benchmark(s)	MA.912.GR.2.1, MA.912.GR.2.3
	The sequence of transformations that will map $\triangle TMV$ onto $\triangle XLN$ is first ○ translate $\triangle TMV$ left 11, up 1, ○ reflect $\triangle TMV$ across the y-axis, ○ rotate $\triangle TMV$ 90° clockwise about the origin, ● rotate $\triangle TMV$ 90° counterclockwise about the origin, then ○ translate $\triangle T'M'V'$ left 3, up 2. ○ translate $\triangle T'M'V'$ left 2, down 3. ● translate $\triangle T'M'V'$ left 6, down 4. ○ reflect $\triangle T'M'V'$ across the $y = -2$. ○ rotate $\triangle TMV$ 90° counterclockwise about the origin.	Recommended Scoring (8 Total Points)	Suggested points are 4 points for each correct answer choice.
Question	5b	Benchmark(s)	MA.912.GR.2.2, MA.912.GR.2.6
	Vertex M maps to vertex <u> L </u>, vertex T maps to vertex <u> X </u>, and vertex V maps to vertex <u> N </u>. Therefore, $\triangle TMV$ is congruent to $\triangle$<u> XLN </u> by the definition of congruence in terms of rigid motions.	Recommended Scoring (8 Total Points)	Suggested points are 4 points for correctly matching all three vertices and 4 points for naming the correct congruent triangle.

Question	6a	Benchmark(s)	MA.912.GR.2.2						
<i>Answers will vary: Side length is preserved because the transformations are rigid motions and rigid motions preserve distance.</i>		Recommended Scoring (4 Total Points)	Consider partial credit for students who correctly note distance is preserved because the transformations are rigid motions but also include incorrect information.						
Question	6b	Benchmark(s)	MA.912.GR.2.1						
<table><tr><th>Mapping</th><th>Algebraic Description</th></tr><tr><td>$\triangle NWG \rightarrow \triangle N'W'G'$</td><td>$(x, y) \rightarrow (-x, -y)$</td></tr><tr><td>$\triangle N'W'G' \rightarrow \triangle N''W''G''$</td><td>$(x, y) \rightarrow (x + 5, y + 1)$</td></tr></table>		Mapping	Algebraic Description	$\triangle NWG \rightarrow \triangle N'W'G'$	$(x, y) \rightarrow (-x, -y)$	$\triangle N'W'G' \rightarrow \triangle N''W''G''$	$(x, y) \rightarrow (x + 5, y + 1)$	Recommended Scoring (8 Total Points)	Suggested points are 4 points for each correct algebraic description.
		Mapping	Algebraic Description						
		$\triangle NWG \rightarrow \triangle N'W'G'$	$(x, y) \rightarrow (-x, -y)$						
		$\triangle N'W'G' \rightarrow \triangle N''W''G''$	$(x, y) \rightarrow (x + 5, y + 1)$						
Additional Notes	If a student incorrectly names the first transformation but correctly names the second transformation using their incorrect naming of the first then the student has correctly completed 4 out of the 8 points.								

Question		7a	Benchmark(s)	MA.912.GR.2.1															
Algebraic Description $(x,y) \rightarrow (x + 4,y - 8)$			Recommended Scoring (4 Total Points)	Consider giving partial credit for students who correctly wrote one of the algebraic descriptions.															
Question		7b	Benchmark(s)	MA.912.GR.2.1															
<table><thead><tr><th>Vertex of <i>WYDM</i></th><th>Work to determine the <i>x</i>-coordinate of <i>HCXF</i></th><th>Work to determine the <i>y</i>-coordinate of <i>HCXF</i></th></tr></thead><tbody><tr><td><i>W</i></td><td>$4 + 4 = 8$</td><td>$2 - 8 = -6$</td></tr><tr><td><i>Y</i></td><td>$5 + 4 = 9$</td><td>$6 - 8 = -2$</td></tr><tr><td><i>D</i></td><td>$8 + 4 = 12$</td><td>$5 - 8 = -3$</td></tr><tr><td><i>M</i></td><td>$9 + 4 = 13$</td><td>$1 - 8 = -7$</td></tr></tbody></table>			Vertex of <i>WYDM</i>	Work to determine the <i>x</i> -coordinate of <i>HCXF</i>	Work to determine the <i>y</i> -coordinate of <i>HCXF</i>	<i>W</i>	$4 + 4 = 8$	$2 - 8 = -6$	<i>Y</i>	$5 + 4 = 9$	$6 - 8 = -2$	<i>D</i>	$8 + 4 = 12$	$5 - 8 = -3$	<i>M</i>	$9 + 4 = 13$	$1 - 8 = -7$	Recommended Scoring (32 Total Points)	Each coordinate for each vertex is worth 4 points. Because students could just copy the vertex values for quadrilateral <i>HCXF</i> the students should show their work.
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			Additional Notes	This question specially addresses clarification 1 of the benchmark.															